A small company sells its chocolates in two different boxes. In the red boxes there are 8 pieces of dark chocolate and 4 pieces of milk chocolate, while in the blue boxes one can find 8 pieces of dark chocolate and 5 pieces of milk chocolate. How many red and blue boxes were produced when they used a total of 336 pieces of dark and 190 pieces of milk chocolate?

red boxes: 20

blue boxes: 22

$$\begin{cases} 8x+8y=336 \\ 4x+5y=190 \end{cases}$$

$$\begin{array}{r} 4x+5y=190 \\ \underline{-)\,4x+4y=168} \\ y=22 \end{array}$$

$x+22=42$

$x=20$

Determine the value of α, when we know that the system $Ax=b$ has infinitely many solutions.

$$A=\begin{pmatrix} 1 & 1 & 2 \\ -1 & -2 & -3 \\ 6 & 8 & 14 \end{pmatrix}, \quad b=\begin{pmatrix} -3 \\ 3 \\ \alpha \end{pmatrix}$$

Answer: -18

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & -3 \\ -1 & -2 & -3 & 3 \\ 6 & 8 & 14 & a \end{array}\right) \begin{array}{c} \\ \mathrm{II}+\mathrm{I} \\ \mathrm{III}-6\mathrm{I} \end{array} = \left(\begin{array}{ccc|c} 1 & 1 & 2 & -3 \\ 0 & -1 & -1 & 0 \\ 0 & 2 & 2 & a+18 \end{array}\right) \begin{array}{c} \\ \\ \mathrm{III}+2\mathrm{II} \end{array} = \left(\begin{array}{ccc|c} 1 & 1 & 2 & -3 \\ 0 & -1 & -1 & 0 \\ 0 & 0 & 0 & a+18 \end{array}\right)$$

$\downarrow$

$a+18=0$

$a=-18$

Using Gaussian elimination we transformed the augmented matrix of the system $Ax=b$ into the form given below, where symbols $*$ stand for the nonzero entries. Mark the true statements.

$$\left(\begin{array}{ccc|c} * & * & * & * \\ 0 & * & * & * \\ 0 & 0 & * & * \\ 0 & 0 & 0 & 0 \end{array}\right)$$

- a. The system is inconsistent.
- b. The system is uniquely solvable.
- c. The vector b is in the subspace generated by the column vectors of A.
- d. The rank of A is 4.
- e. The columns of A are linearly dependent.
- f. The rank of the augmented matrix is 3.
- g. The system has infinitely many solutions.
- h. The rank of A is 3.
- i. The rank of the augmented matrix is 4.
- j. The vector b is not in the subspace generated by the column vectors of A.
- k. The columns of A are linearly independent.

$$\left.\begin{array}{l} \text{rank}(A) = 3 \\ \text{rank}(A|b) = 3 \\ \text{column } A = 3 \end{array}\right\} \text{1 solution}$$

rank (A) = rank(A|b) subspace

rank (A) = column A Independent

In the following system the vector b is not given (but it has a proper size).

$$\begin{pmatrix} 1 & 2 \\ -1 & 0 \\ 1 & 4 \end{pmatrix}_{3\times2} x=b$$

Mark the true statements.

$$\left.\begin{array}{l} \text{rank}(A) = 3 \\ \text{rank}(A|b) = 2 \end{array}\right\} \text{Inconsistent}$$

column A = 2

b = row = 3

x = column = 2

- a. The vector b has 3 elements.
- b. The system has infinitely many solutions for arbitrary value of b.
- c. The vector x has 3 elements.
- d. The vector b has 2 elements.
- e. The system has a unique solution for arbitrary value of b.
- f. The vector x has 2 elements.
- g. There exists a vector b that the system has a unique solution.
- h. There exists a vector b that the system is incinsistent.

Applying Gaussian elimination for the augmented matrix of system $Ax=b$ we obtained the following matrix:

$$\left(\begin{array}{ccc|c} -1 & 1 & 5 & -1 \\ 0 & 1 & 3 & -2 \\ 0 & 0 & -4 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

Mark the true statements.

$$\left.\begin{array}{l} \text{rank}(A) = 3 \\ \text{rank}(A|b) = 3 \\ \text{column } A = 3 \end{array}\right\} \text{1 solution}$$

rank (A) = rank(A|b) subspace

rank (A) = column A independent

- a. The system has a unique solution.
- b. The vector b is not in the subspace generated by the columns of A.
- c. The rank of A is 3.
- d. The system has infinitely many solutions.
- e. The columns of A are linearly dependent.
- f. The vector b is in the subspace generated by the columns of A.
- g. The columns of A are linearly independent.
- h. The rank of A is 4.
- i. The rank of the augmented matrix is 4.
- j. The rank of the augmented matrix is 3.
- k. The system is inconsistent.

Solve the system $Ax=b$, when

$$A=\begin{pmatrix} 1 & 2 & -1 \\ -2 & -7 & 1 \\ -1 & 7 & 6 \end{pmatrix}, \quad b=\begin{pmatrix} -2 \\ -1 \\ 21 \end{pmatrix}$$

$x_1 = -2$

$x_2 = 1$

$x_3 = 2$

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & -2 \\ -2 & -7 & 1 & -1 \\ -1 & 7 & 6 & 21 \end{array}\right) \begin{array}{c} \\ \mathrm{II}+2\mathrm{I} \\ \mathrm{III}+\mathrm{I} \end{array} = \left(\begin{array}{ccc|c} 1 & 2 & -1 & -2 \\ 0 & -3 & -1 & -5 \\ 0 & 9 & 5 & 19 \end{array}\right) \begin{array}{c} \\ \\ \mathrm{III}+3\mathrm{II} \end{array} = \left(\begin{array}{ccc|c} 1 & 2 & -1 & -2 \\ 0 & -3 & -1 & -5 \\ 0 & 0 & 2 & 4 \end{array}\right)$$

$\downarrow$

$2z=4$

$z=2$

$-3y-2=-5$

$-3y=-3$

$y=1$

$x+2-2=-2$

$x=-2$

Mark the invertible matrices.

Select one or more:

a. $\begin{pmatrix} 9 & 7 & 1 \\ 0 & 1 & 5 \\ 0 & 0 & 0 \end{pmatrix}$

b. $\begin{pmatrix} -2 & 1 \\ -8 & 5 \end{pmatrix}$

c. $\begin{pmatrix} 3 & 10 \\ 9 & 31 \end{pmatrix}$

d. $\begin{pmatrix} 8 & -7 \\ 24 & -20 \end{pmatrix}$

e. $\begin{pmatrix} -1 & 1 \\ -6 & 6 \end{pmatrix}$

f. $\begin{pmatrix} -2 & 1 \\ -10 & 5 \end{pmatrix}$

g. $\begin{pmatrix} -6 & 10 \\ 24 & -40 \end{pmatrix}$

h. $\begin{pmatrix} 8 & 8 & 6 \\ -2 & 0 & 1 \\ 0 & 0 & 3 \end{pmatrix}$

a) $\left|\begin{array}{ccc|cc} 9 & 7 & 1 & 9 & 7 \\ 0 & 1 & 5 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right| = 0$

b) $(-10) - (-8) \neq 0$

c) $93 - 90 \neq 0$

d) $(-160) - (-168) \neq 0$

e) $(-6) - (-6) = 0$

f) $(-10) - (-10) = 0$

g) $240 - 240 = 0$

h) $\left|\begin{array}{ccc|cc} 8 & 8 & 6 & 8 & 8 \\ -2 & 0 & 1 & -2 & 0 \\ 0 & 0 & 3 & 0 & 0 \end{array}\right| \neq 0$

In the following system the vector b is not given (but it has a proper size).

$$\begin{pmatrix} 1 & 2 & -2 & 3 \\ -1 & 0 & 1 & 1 \\ 1 & 4 & -2 & -1 \end{pmatrix} x = b$$

Mark the true statements.

- ☒ a. The vector b has 4 elements.
- ☒ ✓ b. The system has infinitely many solutions for arbitrary value of b.
- ✓ c. The vector b has 3 elements.
- ☒ d. The system has a unique solution for arbitrary value of b.
- ☒ e. The vector x has 3 elements.
- ☒ f. There exists a vector b, that the system has a unique solution.
- ☒ ✗ g. There exists a vector b, that the system is inconsistent.
- ✓ h. The vector x has 4 elements.

$\left.\begin{array}{l} \text{rank}(A) = 3 \\ \text{rank}(A|b) = \cancel{4}\ 3 \end{array}\right\}$ no

column A = 4

rank(A) ≠ rank(A|b)   b not in subspace

rank(A) ≠ column A   dependent

b = rows = 3

x = colums = 4

Determine the value of α, when we know that the system $Ax = b$ has infinitely many solutions.

$$A = \begin{pmatrix} 1 & -1 & 1 \\ 3 & -1 & 1 \\ -2 & -2 & 2 \end{pmatrix}, \quad b = \begin{pmatrix} 2 \\ 2 \\ \alpha \end{pmatrix}$$

Answer: −8   4

$$\left(\begin{array}{ccc|c} 1 & -1 & 1 & 2 \\ 3 & -1 & 1 & 2 \\ -2 & -2 & 2 & a \end{array}\right) \begin{array}{l} \text{II}-3\text{I} \\ \text{III}+2\text{I} \end{array} = \left(\begin{array}{ccc|c} 1 & -1 & 1 & 2 \\ 0 & 2 & -2 & -4 \\ 0 & -4 & 4 & a+4 \end{array}\right) \text{III}+2\text{II} = \left(\begin{array}{ccc|c} 1 & -1 & 1 & 2 \\ 0 & 2 & -2 & -4 \\ 0 & 0 & 0 & \cancel{a+8} \end{array}\right)$$

$a+4-8=0$ $a+8=0$

$a-4=0$ $a=-8$

$a=4$

Using Gaussian elimination we transformed the augmented matrix of the system $Ax = b$ into the form given below, where symbols $*$ stand for the nonzero entries. Mark the true statements.

$$\left(\begin{array}{cccc|c} * & * & * & * & * \\ 0 & * & * & * & * \\ 0 & 0 & * & 0 & * \end{array}\right)$$

- ☒ a. The system is uniquely solvable.
- ✓ b. The vector b is in the subspace generated by the column vectors of A.
- ☒ c. The system is inconsistent.
- ✓ d. The rank of A is 3.
- ✓ e. The rank of the augmented matrix is 3.
- ✓ f. The system has infinitely many solutions.
- ✓ g. The columns of A are linearly dependent.
- ☒ h. The vector b is not in the subspace generated by the column vectors of A.
- ☒ i. The rank of the augmented matrix is 4.
- ☒ j. The columns of A are linearly independent.
- ☒ k. The rank of A is 4.

$\left.\begin{array}{l} \text{rank}(A) = 3 \\ \text{rank}(A|b) = 3 \\ \text{column } A = 4 \end{array}\right\} \infty$

rank(A) = rank(A|b)   subspace

rank(A) ≠ column A   dependent

A small company sells its chocolates in two different boxes. In the red boxes there are 8 pieces of dark chocolate and 4 pieces of milk chocolate, while in the blue boxes one can find 8 pieces of dark chocolate and 8 pieces of milk chocolate. How many red and blue boxes were produced when they used a total of 288 pieces of dark and 208 pieces of milk chocolate?

red boxes: 20

blue boxes: 16

$$\begin{cases} 8x + 8y = 288 \\ 4x + 8y = 208 \end{cases}$$

$x + y = 36$

$-)\ x + 2y = 52$

$-y = -16$

$y = 16$

$x + 16 = 36$

$x = 20$

Solve the system $Ax = b$, when

$$A = \begin{pmatrix} 1 & 2 & -1 \\ -2 & -10 & 0 \\ -1 & 4 & 5 \end{pmatrix}, \quad b = \begin{pmatrix} 0 \\ -10 \\ 14 \end{pmatrix}$$

$x_1 =$ 0

$x_2 =$ 1

$x_3 =$ 2

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ -2 & -10 & 0 & -10 \\ -1 & 4 & 5 & 14 \end{array}\right) \begin{array}{l} \text{II}+2\text{I} \\ \text{III}+\text{I} \end{array} = \left(\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 0 & -6 & -2 & -10 \\ 0 & 6 & 4 & 14 \end{array}\right) \text{III}+\text{II} = \left(\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 0 & -6 & -2 & -10 \\ 0 & 0 & 2 & 4 \end{array}\right)$$

$2z = 4$

$z = 2$

$-6y - 4 = -10$

$-6y = -6$

$y = 1$

$x + 2 - 2 = 0$

$x = 0$

Mark the invertible matrices.

Select one or more:

✗ a. $\begin{pmatrix} -3 & -1 & -7 \\ 0 & -9 & 4 \\ 0 & 0 & 0 \end{pmatrix}$

✓ b. $\begin{pmatrix} -7 & 3 \\ -35 & 16 \end{pmatrix}$

✗ c. $\begin{pmatrix} 10 & 4 \\ 50 & 20 \end{pmatrix}$

✓ d. $\begin{pmatrix} -4 & 4 & -9 \\ -7 & 0 & -4 \\ 0 & 0 & -1 \end{pmatrix}$

✗ e. $\begin{pmatrix} -10 & 6 \\ -30 & 18 \end{pmatrix}$

✓ f. $\begin{pmatrix} 3 & 5 \\ 9 & 16 \end{pmatrix}$

✗ g. $\begin{pmatrix} 9 & 7 \\ -18 & -14 \end{pmatrix}$

✓ h. $\begin{pmatrix} -10 & -10 \\ 50 & 51 \end{pmatrix}$

a) $\left|\begin{array}{ccc|cc} -3 & -1 & -7 & -3 & -1 \\ 0 & -9 & 4 & 0 & -9 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right. = 0$

b) $(-112) - (-105) \neq 0$

c) $200 - 200 = 0$

d) $\left|\begin{array}{ccc|cc} -4 & 4 & -9 & -4 & 4 \\ -7 & 0 & -4 & -7 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{array}\right. \neq 0$

e) $(-180) - (-180) = 0$

f) $(48) - (45) \neq 0$

g) $(-126) - (-126) = 0$

h) $(-510) - (-500) \neq 0$

Applying Gaussian elimination for the augmented matrix of system $Ax = b$ we obtained the following matrix:

$$\left(\begin{array}{ccc|c} -2 & 1 & 2 & 0 \\ 0 & -1 & 1 & 2 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

Mark the true statements.

✓ a. The rank of the augmented matrix is 3.

✓ b. The system is inconsistent.

✗ c. The vector b is in the subspace generated by the columns of A.

✗ d. The system has a unique solution.

✓ e. The vector b is not in the subspace generated by the columns of A .

✓ f. The rank of A is 2.

✗ g. The rank of the augmented matrix is 2.

✗ h. The system has infinitely many solutions.

✗ i. The rank of A is 3.

✗ j. The columns of A are linearly independent.

✓ k. The columns of A are linearly dependent.

$\left.\begin{array}{l} \text{rank}(A) = 2 \\ \text{rank}(A|b) = 3 \end{array}\right\}$ Inconsistant

column A = 3

rank(A) = rank(A|b) ~~subspace~~

rank(A) ≠ column A dependent

Applying Gaussian elimination for the augmented matrix of system $Ax=b$we obtained the following matrix:

$$\left(\begin{array}{ccc|c} 5 & 1 & 2 & -1 \\ 0 & 3 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

$\left.\begin{array}{l} \text{rank}(A) = 2 \\ \text{rank}(A|b) = 2 \\ \text{column } A = 3 \end{array}\right\}$ ∞ solutions

rank(A) = rank(A|b) b is in subspace.

rank(A) ≠ column A dependent

Mark the true statements.

- ✗ a. The system is inconsistent.
- ✓ b. The vector b is in the subspace generated by the columns of A.
- ✗ c. The columns of A are linearly independent.
- ✗ d. The vector b is not in the subspace generated by the columns of A.
- ✗ e. The system has a unique solution.
- ✓ f. The columns of A are linearly dependent.
- ✓ g. The rank of A is 2.
- ✓ h. The system has infinitely many solutions.
- ✗ i. The rank of the augmented matrix is 3.
- ✓ j. The rank of the augmented matrix is 2.
- ✗ k. The rank of A is 3.

Mark the invertible matrices.

Select one or more:

- ✗ a. $\begin{pmatrix} 6 & 5 \\ -36 & -30 \end{pmatrix}$
- ✓ b. $\begin{pmatrix} 7 & -3 \\ 21 & -8 \end{pmatrix}$
- ✗ c. $\begin{pmatrix} 5 & 6 \\ -25 & -30 \end{pmatrix}$
- ✗ d. $\begin{pmatrix} -8 & 3 \\ 16 & -6 \end{pmatrix}$
- ✓ e. $\begin{pmatrix} 3 & 9 \\ 6 & 19 \end{pmatrix}$
- ✗ f. $\begin{pmatrix} -9 & 3 & 5 \\ 0 & 1 & -10 \\ 0 & 0 & 0 \end{pmatrix}$
- ✓ g. $\begin{pmatrix} 3 & 4 \\ 12 & 17 \end{pmatrix}$
- ✓ h. $\begin{pmatrix} 9 & -10 & -10 \\ -8 & 0 & -7 \\ 0 & 0 & -5 \end{pmatrix}$

a) $(-180)-(-180)=0$

b) $(-56)-(-63) \neq 0$

c) $(-150)-(-150)=0$

d) $(48)-(48)=0$

e) $(57)-(54) \neq 0$

f) $\left|\begin{array}{ccc|cc} -9 & 3 & 5 & -9 & 3 \\ 0 & 1 & -10 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right|$ $0+0+0$

g) $(51)-(48) \neq 0$

h) $\left|\begin{array}{ccc|cc} 9 & -10 & -10 & 9 & -10 \\ -8 & 0 & -7 & -8 & 0 \\ 0 & 0 & -5 & 0 & 0 \end{array}\right|$ $0+0+0$ $\neq 0$

Solve the system $Ax=b$, when

$$A=\begin{pmatrix} 1 & 2 & -1 \\ -1 & 2 & -1 \\ -1 & 0 & 2 \end{pmatrix}, \quad b=\begin{pmatrix} -2 \\ -6 \\ 2 \end{pmatrix}$$

$x_1 =$ 2

$x_2 =$ -1

$x_3 =$ 2

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & -2 \\ -1 & 2 & -1 & -6 \\ -1 & 0 & 2 & 2 \end{array}\right) \begin{array}{l} \\ II+I \\ III+I \end{array} = \left(\begin{array}{ccc|c} 1 & 2 & -1 & -2 \\ 0 & 4 & -2 & -8 \\ 0 & 2 & 1 & 0 \end{array}\right) \begin{array}{l} \\ \\ III-\frac{1}{2}II \end{array} = \left(\begin{array}{ccc|c} 1 & 2 & -1 & -2 \\ 0 & 4 & -2 & -8 \\ 0 & 0 & 2 & 4 \end{array}\right)$$

$2z=4$

$z=2$

$4y-2z=-8$

$4y-4=-8$

$4y=-4$

$y=-1$

$x+2y-z=-2$

$x-2-2=-2$

$x-4=-2$

$x=2$

In the following system the vector b is not given (but it has a proper size).

$$\begin{pmatrix} 1 & 2 & -2 & 3 \\ -1 & -2 & 2 & 1 \\ 2 & 4 & -4 & -1 \end{pmatrix} x = b$$

$\left.\begin{array}{l} \text{rank}(A) = 3 \\ \text{rank}(A|b) = \cancel{3}\,2 \\ \text{column } A = 4 \end{array}\right\}$ ~~∞ solutions~~ inconsistant

rank(A) = rank(A|b) → b is in subspace

rank(A) ≠ column A → dependent

b = raw = 3

x = colum = 4

Mark the true statements.

- ✓ a. There exists a vector b that the system is inconsistent.
- ✗ b. There exists a vector b that the system has a unique solution.
- ✗ c. The system has a unique solution for arbitrary value of b.
- ✗ d. The vector x has 3 elements.
- ✗ e. The system has infinitely many solutions for arbitrary values of b.
- ✗ f. The vector b has 4 elements.
- ✓ g. The vector x has 4 elements.
- ✓ h. The vector b has 3 elements.

Using Gaussian elimination we transformed the augmented matrix of the system $Ax=b$into the form given below, where symbols $*$ stand for the nonzero entries. Mark the true statements.

$$\left(\begin{array}{ccc|c} * & * & * & * \\ 0 & * & * & * \\ 0 & 0 & * & 0 \end{array}\right)$$

$\left.\begin{array}{l} \text{rank}(A) = 3 \\ \text{rank}(A|b) = 3 \\ \text{column } A = 3 \end{array}\right\}$ 1 solution

rank(A) = rank(A|b) → b is in subspace

rank(A) = column A → independent

- ✓ a. The columns of A are linearly independent.
- ✗ b. The system is inconsistent.
- ✗ c. The rank of the augmented matrix is 4.
- ✗ d. The rank of A is 4.
- ✓ e. The rank of A is 3.
- ✗ f. The system has infinitely many solutions.
- ✓ g. The system is uniquely solvable.
- ✓ h. The rank of the augmented matrix is 3.
- ✗ i. The columns of A are linearly dependent.
- ✗ j. The vector b is not in the subspace generated by the column vectors of A.
- ✓ k. The vector b is in the subspace generated by the column vectors of A.

A small company sells its chocolates in two different boxes. In the red boxes there are 8 pieces of dark chocolate and 4 pieces of milk chocolate, while in the blue boxes one can find 10 pieces of dark chocolate and 7 pieces of milk chocolate. How many red and blue boxes were produced when they used a total of 208 pieces of dark and 128 pieces of milk chocolate?

red boxes: 11

blue boxes: 12

$$\begin{cases} 8x + 10y = 208 \\ 4x + 7y = 128 \end{cases}$$

$$\begin{aligned} 4x + 5y &= 104 \\ -)\ 4x + 7y &= 128 \\ \hline -2y &= -24 \\ y &= 12 \\ 4x + 60 &= 104 \\ 4x &= 44 \\ x &= 11 \end{aligned}$$

Determine the value of α, when we know that the system $Ax = b$ has infinitely many solutions.

$$A = \begin{pmatrix} 1 & -3 & -3 \\ -3 & 12 & 10 \\ 11 & -42 & -36 \end{pmatrix}, \quad b = \begin{pmatrix} -4 \\ 3 \\ \alpha \end{pmatrix}$$

Answer: -71 -17

$$\left(\begin{array}{ccc|c} 1 & -3 & -3 & -4 \\ -3 & 12 & 10 & 3 \\ 11 & -42 & -36 & a \end{array}\right) \begin{array}{l} \mathrm{II}+3\mathrm{I} \\ \mathrm{III}-11\mathrm{I} \end{array} = \left(\begin{array}{ccc|c} 1 & -3 & -3 & -4 \\ 0 & 3 & 1 & -9 \\ 0 & 9 & -3 & a+44 \end{array}\right) \mathrm{III}-3\mathrm{II} = \left(\begin{array}{ccc|c} 1 & -3 & -3 & -4 \\ 0 & 3 & 1 & -9 \\ 0 & 0 & -6 & a+71 \end{array}\right)$$

$a = -71$

$$\left(\begin{array}{ccc|c} 1 & -3 & -3 & -4 \\ -3 & 12 & 10 & 3 \\ 11 & -42 & -36 & a \end{array}\right) \begin{array}{l} \mathrm{II}+3\mathrm{I} \\ \mathrm{III}-11\mathrm{I} \end{array} = \left(\begin{array}{ccc|c} 1 & -3 & -3 & -4 \\ 0 & 3 & 1 & -9 \\ 0 & -9 & -3 & a+44 \end{array}\right) \mathrm{III}+3\mathrm{II} = \begin{pmatrix} 1 & -3 & -3 & 4 \\ 0 & 3 & 1 & -9 \\ 0 & 0 & 0 & a+32 \end{pmatrix}$$